

Ali Momennasab

amomennasab@ucsd.edu | alimomennasab.github.io | github.com/alimomennasab | (626) 393-8922

Education

University of California, San Diego

9/2026 - 12/2027 (Expected)

Master of Science, Computer Science | Artificial Intelligence Specialization

California State Polytechnic University, Pomona

8/2022 - 1/2026

Bachelor of Science, Computer Science, Minor in Data Science | **3.93/4.0 GPA**

- Relevant Coursework: Machine Learning, Computer Vision, Generative AI, GPU Computing

Experience

Research Intern | University of California, San Diego

7/2026 - Present

Researching latent reasoning and one-step image generation under Professor Lianhui Qin.

- Implemented MeanFlow and drifting models with Diffusion Transformer (DiT) backbones in PyTorch, achieving high-quality generation in a single inference step.
- Improved latent MeanFlow training with Muon optimization and perceptual reconstruction loss, speeding up convergence and enhancing image quality.

Research Assistant | University of California, San Francisco

6/2023 - 6/2026

Researching deep learning segmentation of cardiac ultrasounds under Professor Jan Christoph.

- Developed a PyTorch temporal 3D U-Net for cardiac ultrasound segmentation with residual/dense block variants and on-the-fly augmentations, improving F1 scores from 0.79 to 0.92.
- Trained conditional GANs and diffusion models to generate synthetic ultrasound volumes from segmentation masks, achieving Dice scores of 0.88 between conditioning masks and segmentations of generated volumes.

Research Assistant | California State Polytechnic University, Pomona

1/2025 - 3/2025

Researching deep learning survival analysis of whole-slide cancer images under Professor Sai Kosaraju.

- Built a multimodal CNN for survival prediction using pathology images and genomic data, achieving C-indices of 0.95 (brain) and 0.73 (lung); published at [ACM BCB 2026](#).
- Developed a hybrid Vision Transformer and GNN framework for whole-slide image embedding and tumor subtype clustering, achieving 0.82 F1.

Instructor | Code Ninjas Hacienda Heights

6/2023 - 8/2024

- Taught 1-on-1 programming lessons and led web development camps for K-12 students using Unity, Roblox Studio, HTML/CSS, and JavaScript.

Projects

ASL Word & Letter Translator | github.com/alimomennasab/ASL-Translator

- Developed and trained 2D CNNs for static ASL letter classification (95% accuracy), and 3D CNNs for video-based word recognition (40% accuracy).
- Built a MediaPipe hand detection and augmentation pipeline, expanding the dataset from 500 to 5,000 samples and doubling word recognition accuracy from 20% to 40%.

Generative Music Genre Transfer | github.com/alimomennasab/CS4990-Generative-AI

- Implemented VAE, GAN, and WGAN models in PyTorch for symbolic music genre transfer (jazz/pop/classical), observing greater training stability and reconstruction consistency with VAEs.
- Designed a MIDI file tokenization script, converting 2,220 songs into 13 million training tokens.

Skills

Languages/Libraries: Python, C++, Java, CUDA, scikit-learn, NumPy, pandas, TypeScript/JavaScript

Frameworks: PyTorch, TensorFlow, Pinecone, React, Next.js, Express

Developer Tools: Git, Docker, AWS, PostgreSQL, MongoDB